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Are postgraduate programs enabling technological and instructional leadership among future school leaders? An institutional case study

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Abstract

It is important that higher education institutions equip future school leaders with a holistic view of teachers' digital competence. This qualitative study explores the conceptual understanding of teacher digital competence among future school leaders in Spain, comparing their perspectives with the DigCompEdu framework. A total of 41 students enrolled in a master's programme in school management participated, responding to a single open question about the use of technology in teaching practice. Data were analysed using content analysis, comparing the responses with the 22 competences of the DigCompEdu framework. The findings reveal limitations in their understanding of the full scope of teacher digital competence. While participants acknowledged basic uses of technology in teaching, learning and professional engagement, they demonstrated conceptual gaps in areas such as developing students' digital competence, empowering learners and effective digital assessment. These findings highlight the need for a holistic approach to teacher digital competence in leadership training to ensure that future school leaders are prepared for digital and pedagogical leadership. In practice, this means that school leadership training programmes should be reviewed to ensure that they cover all dimensions of teacher digital competence, thereby preparing future leaders to effectively implement and support the integration of technology in their schools.

Keywords Initial leadership training, Instructional leadership, Technological leadership, Digital competence, School leaders

Introduction

Digital technologies have the potential of transforming educational practices in different ways. They can foster active methodologies (Cheng et al., 2019; Prince, 2004), personalize learning (Escueta et al., 2020; Fyfe, 2016; Surma & Kirschner, 2020), facilitate better and timely feedback (Fyfe, 2016; Hattie & Timperley, 2007), improve interaction and collaboration among students (Bernard et al., 2009; Castaño-Muñoz et al., 2014), help to scaffold the learning process (McLaren et al., 2016; Reis-Andersson, 2024), facilitate

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bimodal presentation of content (Moreno & Mayer, 2007) or help to develop students digital competence, among others.

While teachers undoubtedly play a crucial role in the effective incorporation of digital technologies in teaching and learning practices, this also hinges on external factors such as the socioeconomic context, the existing conditions inside the school, and the support offered by school leaders (Lucas et al., 2021; Perrotta, 2013; Timotheou et al., 2023; Vanderlinde & van Braak, 2010).

School leaders are important agents of institutional change (Acton, 2021). Studies have identified multiple functions that school leaders must perform to effectively lead and manage change within their institutions (Pont et al., 2008). These functions encompass (co)developing a strategic view for the school and translating it into actionable improvement plans that can be monitored (Day et al., 2020); overseeing resource management, including budgetary administration and maintenance of school infrastructures; managing human capital by coordinating staff responsibilities and fostering school collaboration and continuous professional development (Radinger, 2014; Reis-Andersson, 2024); nurturing a positive school culture and climate by promoting well-being and shared goals (Velarde et al., 2022); representing the school in the relationships with stakeholders (e.g., administration, other schools, associations, families); and building partnership and networks with other entities (Valli et al., 2018).

Moreover, in the last decades, new functions have been added, and school leadership has shifted from a managerial focus to instructional and curriculum leadership, where pedagogical innovation and teacher support are more important elements for the attainment of organisational change (Glatthorn & Jailall, 2009). In addition, recent leadership theories highlight the need for current school leaders to become technological leaders (Anderson & Dexter, 2005; Chang, 2012), that is, school leaders who, in addition to the elements mentioned above, are also knowledgeable about the transformative potential of digital technologies and their potential pitfalls to manage technological change (Pettersson, 2018). From this perspective, technology is not only a tool, but a fundamental element in the work of school leaders, as it directly influences management, decision making and improvement of educational processes (Krein, 2025).

Within this modern leadership paradigm, the responsibilities of school leaders extend to the realm of digital transformation, where their influence manifests in diverse ways. They are called upon to articulate a vision and strategy for the integration of technology into the school environment (Costa et al., 2021), creating beneficial conditions and environments fostering innovation-supporting ecosystems that nurture technological advancements. Leaders must ensure the upkeep and maintenance of connections and technical infrastructures, laying the foundation for seamless digital learning experiences (Hrustek et al., 2023). They should encourage and facilitate collaboration, prioritizing time for digital leadership (Håkansson Lindqvist, 2019), fostering the sharing of digital resources and the exchange of effective pedagogical technology integration practices (Castaño Muñoz et al., 2021). Furthermore, Tondeur et al. (2010) and Avidov-Ungar et al. (2022), highlight the importance of digital coordinators, where such a role exists, and their coordination with school leaders for an effective ICT integration. To achieve widespread digital education, school leaders also play an important role in building an appropriate culture and consensus among the entire school community, responding to different agents' worries about the integration of technologies in schools and aligning

their different interests in this topic (Blau & Shamir-Inbal, 2017). Leaders must also plan and execute professional development programs for teachers, equipping them with the skills and knowledge required to effectively integrate technology into their teaching practices (Tondeur et al., 2016) and promote innovative teaching and learning (Håkansson Lindqvist, 2019). Finally, school leaders should provide technical and pedagogical support, offering guidance and assistance to teachers who may require additional support in effectively incorporating digital technologies into their teaching practices (Chang, 2012; Håkansson Lindqvist, 2019).

Digital transformation in schools depends on leaders who not only utilize technology for managerial tasks, such as enhancing communication and administrative efficiency (Kafa, 2025) but also possess a deep understanding of how technology can improve teachers' professional practices (Makri & Vlachopoulos, 2019). Leaders with strong digital knowledge are more likely to successfully drive digital transformation, fostering innovation and a digital culture. Their confidence and skills in handling digital technologies facilitate the adoption of new tools. Conversely, leaders experiencing digital anxiety may obstruct progress, generating resistance and slowing the transition to digital environments (Chen &Kouhsari, 2025).

However, successful implementation faces significant challenges, including poor infrastructure and, specially, inadequate specific training (Berkovich & Hassan, 2023; Kafa, 2025; Lozano-Díaz et al., 2024). School leaders also highlight a persistent lack of knowledge and the need for continuous support to effectively implement these technologies (Masoumi & Noroozi, 2023; Olofsson et al., 2015). Addressing these challenges requires equipping school leaders with comprehensive knowledge through initial and continuous training programs that go beyond basic technical skills (Afshari et al., 2012; Håkansson Lindqvist, 2019). Universities and educational authorities can play an important role by offering programs that incorporate pedagogical implications (Flanagan & Jacobsen, 2003; Håkansson Lindqvist, 2019; Voogt et al., 2013) and ethical considerations (Selwyn et al., 2020; Williamson et al., 2019) of technology selection and use. This comprehensive approach can empower leaders to provide a mix of technological and pedagogical leadership.

Teacher digital competence

The identification and definition of "what can be done" with digital technologies by teachers in their professional practice has been discussed from several perspectives and, consequently, various frameworks have been proposed to specify it. Lucas et al. (2021) highlight the following frameworks developed with this aim by different organizations: UNESCO ICT Competence Framework for Teachers (UNESCO, 2011); ISTE National Educational Technology Standards (NETS) and Performance Indicators for Teachers (ISTE, 2000); Rubric for Effective Teacher Technology Use (Johnson & Mielke, 2013); the Norwegian Professional Digital Competence Framework for Teachers (Kelentrić et al., 2017); and the TET-SAT, developed by MENTEP (MENTEP, 2018). In addition, they also mention some scientific articles where frameworks in the same direction are proposed (Mishra & Koehler, 2006; Tondeur et al., 2017). The ultimate idea of all these frameworks is to define, in different dimensions or areas, what teaching digital competence is, in order to facilitate its acquisition by teachers and promote innovation associated with the digital transformation of education systems.

While teachers' digital competence was recognized as significant pre-COVID and its emphasis intensified during the pandemic, it remains a social and policy priority worldwide in 2025. In social terms, the European Union, for instance, launched the Digital Education Action Plan 2021–2027 (European Commission, 2020) post-COVID, with targeted actions to enhance both teachers' and school leaders' digital competence across the EU. This plan aims to modernize education through innovation and improve learners' digital skills by prioritizing the development of digital proficiency among educators and leaders alike. Setting goals to be achieved by 2030, the EU demonstrates a long-term strategic commitment to teachers' digital competence and its current relevance. Moreover, the EU recognizes school leaders, alongside teachers, as central agents for successful educational reform, including the digital transformation, and fosters initiatives at the school level. This aligns with the increasing volume and importance n digital leadership research following COVID-19 (Karakose et al., 2024). Recognizing that the quality of both teachers and school leaders directly impacts student outcomes, the EU prioritizes their continuous professional development.

At the European level, in recent years, the European Framework for the Digital Competence of Educators, or DigCompEdu, has been launched (Redecker & Punie, 2017). This framework defines 22 educator-specific digital competences, organised in six areas: "Professional Engagement", "Digital Resources", "Teaching and learning", "Assessment", "Empowering Learners", and "Facilitating Learners' Digital Competence". Each area is composed of a set of specific competences that, together, allow teachers to effectively integrate digital technologies into their teaching methods and nurture the digital competence of their students. According to the DigCompEdu framework (Redecker & Punie, 2017), these areas are defined as follows:

- *Professional engagement* it includes the competences required in the use of digital technologies for: (a) organizational communication, with learners, parents or third parties; (b) professional collaboration, exchanging knowledge and practices or working together on innovative practices; (c) reflective practice on their digital practice and that of their colleagues; and (d) continuous professional development in the use of digital technologies.
- *Digital resources* this area includes competences required in the use of digital technologies for: (a) selecting; (b) creating (or co-creating) and modifying digital resources; and (c) managing, protecting, and sharing digital resources respecting aspects such as privacy and copyright rules.
- *Teaching and learning* it includes competences required in the use of digital technologies for: (a) teaching, improving teaching, and learning strategies via the use of digital technologies for implementing innovative pedagogical methods or orchestrate digital learning; (b) providing better guidance and support to learners; (c) fostering collaborative learning; and (d) supporting self-regulated learning (i.e., enabling learners to plan, monitor and reflect on their own learning).
- *Assessment* this area includes competences required in the use of digital technologies for: (a) enhancing the diversity and suitability of assessment strategies; (b) analysing and interpreting digital evidence on learner activity to inform teaching and learning; and (c) providing feedback and adapting teaching strategies accordingly.
- *Empowering learning* it includes competences required in the use of digital technologies for: (a) accessibility and inclusion, ensuring accessibility to learning

resources and activities for everyone; (b) differentiation and personalization, addressing learners' diverse learning needs, rhythms, levels and goals; and (c) actively engaging learners, fostering learners' active and creative engagement or supporting the development of transversal skills.

– *Facilitating learners´ digital competence* this area includes competences required in the use of digital technologies for developing learners´: (a) information and media literacy; (b) digital communication and collaboration competences; (c) digital content creation; (d) competences for the responsible use of technologies and wellbeing; and (e) competences for digital problem solving, which requires learners to identify and solve technical problems or transfer technological knowledge to new situations.

Aims of the study

In line with instructional and technological leadership theories, school leaders are tasked with fostering the institutional conditions that enable teachers to effectively utilize digital technologies in their teaching practices and providing support and guidance to those teachers who face challenges in developing digital practices.

Research has established a positive relationship between technological leadership and teachers´ effective integration of digital technologies (Bastos & Oliveira, 2015; Berkovich & Hassan, 2023). However, for school leaders to effectively exercise this type of leadership, they must possess a comprehensive understanding of the possibilities that digital technologies offer to teachers. Unfortunately, the literature shows that school leaders often have not been prepared for technological leadership (Lozano-Díaz et al., 2024; Masoumi & Noroozi, 2023). Consequently, they perceive digitalization as a difficult challenge (Olofsson et al., 2015), and exhibit a lack of skills in managing and fostering the digital transformation of teaching practices (Acton, 2021; Agyei & Voogt, 2014; Masoumi & Noroozi, 2023), as well as a limited understanding of all the options that digital technologies offer to teachers.

To contribute to addressing this problem, we conducted a qualitative study to explore the understanding of students enrolled in a master's program track focused on 'School Management' regarding the transformative potential of digital technologies in enhancing teachers' professional practice (i.e., teachers' digital competence).

To further elucidate their perspectives, we conducted a content analysis of the data gathered from participants' responses to a single, open-ended written question, and then compared the resulting categories against the established 22 competences outlined in the DigCompEdu framework.[1] This approach enabled us to identify what competences are commonly recognized and discussed among aspiring school leaders and what competences are often overlooked or underemphasized. In this way, we can pinpoint knowledge gaps and areas for improvement in the understanding of aspiring school leaders regarding the impact and possibilities of digital technologies on teaching and learning practices.

[1] As discussed above, school leaders need a comprehensive vision for ICT integration that must extend beyond just teaching and learning, encompassing all areas of school transformation. Frameworks like DigCompOrg, for instance, include broader domains such as leadership, governance, and collaboration (Kampylis et al. 2015). Nevertheless, from the pedagogical leadership perspective taken in this study it is important that they possess in-depth knowledge of how teachers can utilize ICTs as this understanding is key for providing meaningful support. Consequently, for the purposes of our study, we draw upon the DigCompEdu framework to specifically address the digital competence of educators.

Our findings provide insights for educational stakeholders, particularly for higher education institutions and other entities responsible for designing training programs for school leaders. By understanding the conceptual knowledge gaps, these institutions can identify training needs and, by focusing on them, empower aspiring and practicing school leaders to fully exercise their digital and instructional leadership in an era where school leaders' support is important to navigate the ever-evolving landscape of digital technologies.

Methods

Participants

The participants of this study were 41 students enrolled in a school leadership preparation program, specifically in a one-year master's degree in "Management, Evaluation and Quality of Training Institutions" during the academic year 22/23 or 23/24. This master's program is delivered by the University of Seville (Spain) and offers two specializations: school management or assessment. In our study, we limited participation to students majoring in school management. Subjects in this itinerary encompass topics related to school management, such as legal, formal, and organisational aspects of schools, human resource management, project management, etc. Consequently, in this itinerary, there is an overrepresentation of students with current or future interest in becoming school leaders.

The average age of the participants was 26 years and most of them were females (80%).[2] The majority of the participants had a BA degree related to education (mostly in Primary Education, Pre-primary Education, or Pedagogy) and less than 15% had previously completed another master's degree (Table 1). Regarding their previous experience, 46% of the participants had teaching experience (excluding mandatory degree internships).

Data collection and analysis

All the students enrolled in the school management itinerary of the Master's in any of the two school years in which the data collection was carried out (22/23 and 23/24) were contacted for participation. In both cases, the procedure to inform students about the study was the same. Firstly, a member of the research team explained the study to them

Table 1 Participants´ characteristics and participation rates

	22/23	23/24	Total
Gender			
Male	26.3%	13.6%	19.5%
Female	73.7%	86.4%	80.5%
Age (mean)	25.7	26	25.9
Previous teaching experience	47.3%	45.5%	46.3%
Previous education			
Master	10.5%	18.2%	14.5%
Degree	89.5%	81.8%	85.5%
n enrolled	22	27	49
n participants	19	22	41
% participation	86.4%	81.5%	83.6%

[2] Regarding gender identity, during data collection, participants were given the opportunity to self-identify as it was asked as open question. However, all responses resulted in either male or female identification.

and then an informed consent with all the details about the study and data treatment was provided for signature.

The data collection was performed during class-time, which led to high participation rates and prevented possible selection bias. In the first year, 19 out of 22 students participated in the study, and during the second year, 22 out of 27 students participated in the study. The total number of participants was 41, with a total participation rate of 83.6%. To avoid possible social desirability biases, the anonymity of the responses was guaranteed.

The aim of this study was to gather what the participants understand by educators' digital competence and the dimensions/concepts associated with it, to compare their responses with the theoretical definition (areas and subareas) provided by the European Framework for the Digital Competence of Educators (*DigCompEdu*).

The use of qualitative open single questions has been reported to be a good method for understanding subjective perceptions and how individuals or a specific group of people conceive and value complex concepts (Erikson et al., 2018). Moreover, it has also been identified as an appropriate method to capture the elements that individuals associate with specific concepts in a free and not guided/biased way (Holloway et al., 2008). Furthermore, single-item measures can be deemed acceptable when the construct under evaluation is clearly defined (Fuchs & Diamantopoulos, 2009), as is the case in our study. This methodology offers advantages: it allows for the use of research protocols that are not excessively lengthy and proves more satisfactory for both participants and researchers (Allen et al., 2022). This type of measure is considered as valid and reliable as its multi-item counterparts (Ahmad et al., 2014; Ang & Eisend, 2018).

Students were asked to respond to this single question: 'Taking the perspective of a primary or secondary education teacher[3]: list and reflect on aspects/dimensions where you most consider that the use of ICT can be useful in professional practice'. Although participants are students on a master focused on school management and have different teaching experience levels, understanding their perspectives on how teachers can leverage technology in its practice is important to check if their training have effectively developed their future instructional and technological leadership capabilities.

Methodologically, we employed a broad, open-ended question to investigate participants' spontaneous associations with teachers' digital competence considering our research goals. Although directing participants to specific competence areas could have increased the frequency of related competence mentions, this would have artificially structured their responses. Furthermore, it would have shifted the study's focus, as our primary aim was to explore the cognitive landscape participants naturally employ when considering teacher digital competence, without the influence of prompt.

To avoid confusion and facilitate an accurate data collection, we did not use the term 'digital competence' in the question but a simple operationalization of it: how the use of ICT can be useful in (teachers') professional practice. Moreover, to mitigate potential cognitive load associated with a single, broad open-ended question, we implemented several methodological controls. First, participants were granted a dedicated 30-min period for individual reflection and written response. This extended timeframe was designed to facilitate thorough processing of the question, enabling participants to recall

[3] To contextualize the response, the level was selected by the participants before answering to the single open question.

relevant experiences and formulate detailed, comprehensive, and structured responses regarding the dimensions and associations that come to the participants' minds when thinking on digital competence. Second, data collection occurred during scheduled class time in a controlled, quiet environment, minimizing external distractions and promoting focused engagement. Individual responses, without communication among students, were mandated to ensure participant focus, autonomy and to capture each individual's unique associations with digital technologies in teaching practice. Finally, the written format of the responses served to externalize cognitive processes, reducing working memory demands and fostering more considered and structured articulations compared to verbal modalities.

For the data analysis, the content analysis method was used (Drisko & Maschi, 2016). This methodology follows a deductive logic that allows determining the presence of predefined codes in the analyzed responses. Specifically, the codes used in our study correspond to the areas and subareas of the European Framework of Digital Competence for Educators (DigCompEdu).[4] Three categories were included in the codebook: professional competences of educators, pedagogical competences of educators and Student competences. Each category in turn contained subcategories: "Professional Engagement", "Digital Resources", "Teaching and Learning", "Assessment", "Empowering Learners" and "Facilitating Learners' Digital Competence". A total of 22 codes (Annex 1) were used to code the documents using lines and paragraphs as the unit of analysis.

The presented results are those of the number of participants who mentioned each of the areas or specific competences of the framework. This approximation allows understanding how teacher digital competence was understood by the participants, as well as identifying those competences that are considered important by the DigCompEdu framework, but which future principals do not regard as much. Furthermore, differences between participants with and without professional experience as teachers were explored for robustness check (see Fig. 3), as variations in background might influence their perceptions of digital competences. If no substantial differences were observed between these groups for a particular area, this would indicate a consistent understanding or emphasis on that competence regardless of professional exposure, thereby strengthening the validity of the study´s findings despite the mix of participant pool.

The categorization and analysis process were carried out using ATLAS.ti software v23.3.0. To guarantee the quality and validity of the coding, each category was independently revised and triangulated as the data were coded independently by two members of the research team. The initial observed agreement between codes applied was 81%, which translates to a Cohen's kappa of 0.62. When there were discrepancies, they were discussed until an agreement was reached.

Results

This section presents the main results on the future school principals' understanding of teacher digital competence, employing the DigCompEdu as reference framework.

To understand how future principals perceive teacher digital competence, we analysed how many participants mentioned each DigCompEdu area (Fig. 1). The most cited were "Professional Engagement" and "Teaching and Learning" (26 participants, 63.4%),

[4] The codebook structure is detailed in Annex I, while the European Framework for the Digital Competence of Educators (see Redecker & Punie, 2017) provides comprehensive explanations of each area and competence.

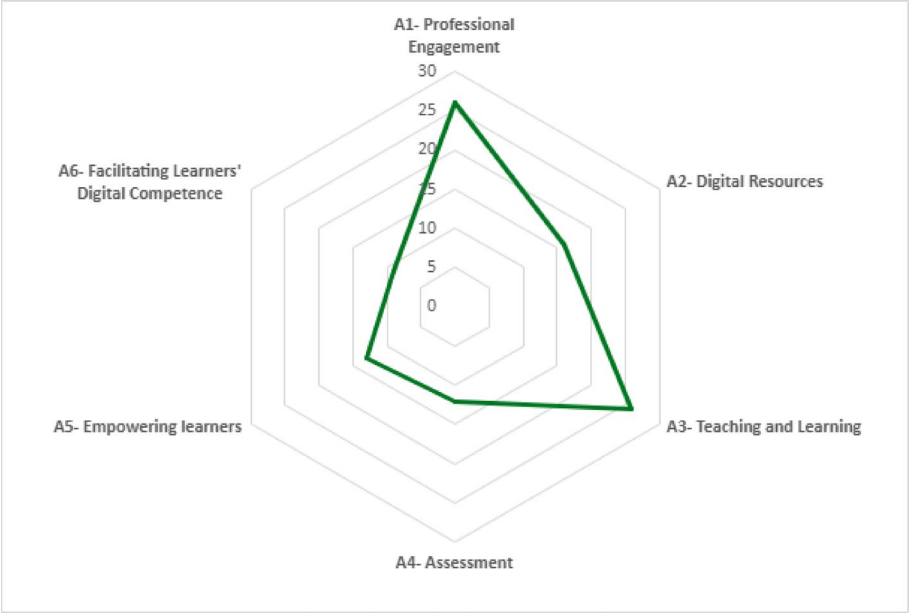


Fig. 1 Number of participants mentioning each area

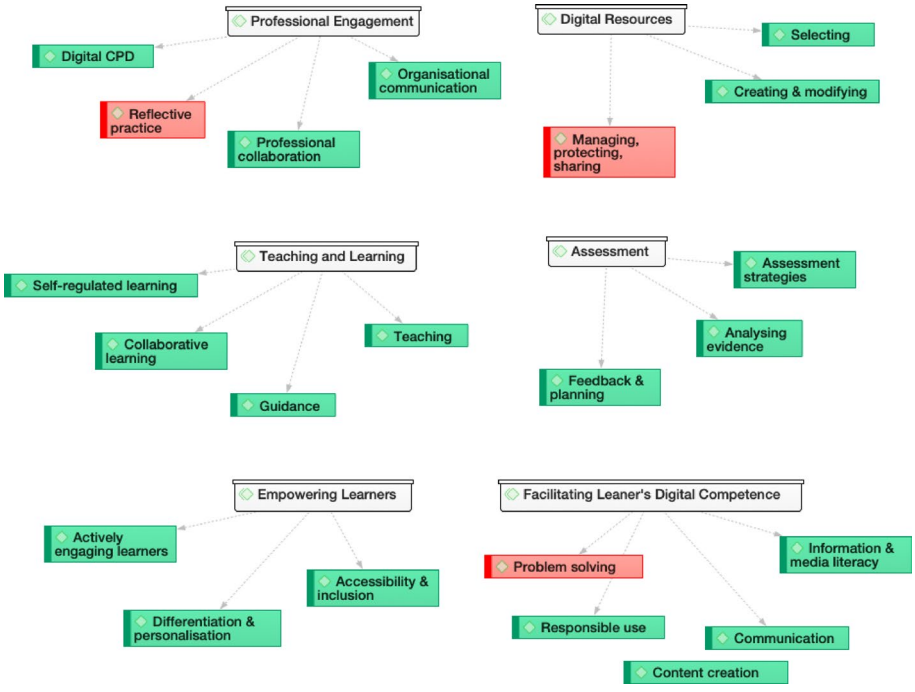


Fig. 2 Competences mentioned at least once (green) and competences omitted (red)

followed by "Digital Resources" (16 mentions, 39%), showing moderate consistency. In contrast, "Empowering Learners" (13 mentions, 31.7%), "Assessment" (12 mentions, 29.2%), and "Facilitating Learners' Digital Competence" (9 mentions, 22%) were the least mentioned (66).

Figure 2 presents a detailed analysis of the specific competences within each area of the DigCompEdu framework, highlighting which were mentioned by participants (green) and which were omitted (red). Competences marked in green indicate areas

where participants demonstrated a more robust recognition and understanding. For example, in "Professional Engagement", "Organisational Communication" was widely recognised, reflecting the importance participants place on effective communication with the educational community. Conversely, competences marked in red indicate areas that were notably overlooked or underestimated by participants. This includes critical aspects such as "Reflective Practice" and "Managing, Protecting and Sharing" in "Digital Resources", as well as "Problem Solving". The omission of these competences suggests potential gaps in the training of future school leaders, which could affect their ability to effectively lead the integration of technology in their institutions."

Figure 3 illustrates the percentage[5] of participants mentioning each area, categorized by their previous professional experience. An overall trend observed is that participants with previous professional experience tend to slightly mention all areas more frequently. However, for most areas (A1-A5), the proportions of participants with and without prior experience do not show important differences, indicating a generally consistent emphasis across these domains regardless of professional experience. However, a divergence is observed in "A6-Facilitating Learners' Digital Competence," where participants with previous experience show a significantly higher percentage compared to those without, suggesting that professional engagement may increase the relevance given to this are more focused on learners´ outcomes than in processes. Given the overall similarity across most categories and the objective of this study to understand broader trends, subsequent analyses will proceed with the two groups combined.

Table 2 presents the number of mentions for each specific competence within the DigCompEdu areas. Analysing this alongside the qualitative data provides a clear picture of how future school leaders understand teacher digital competence. The results are detailed below in descending order of mentions per area.

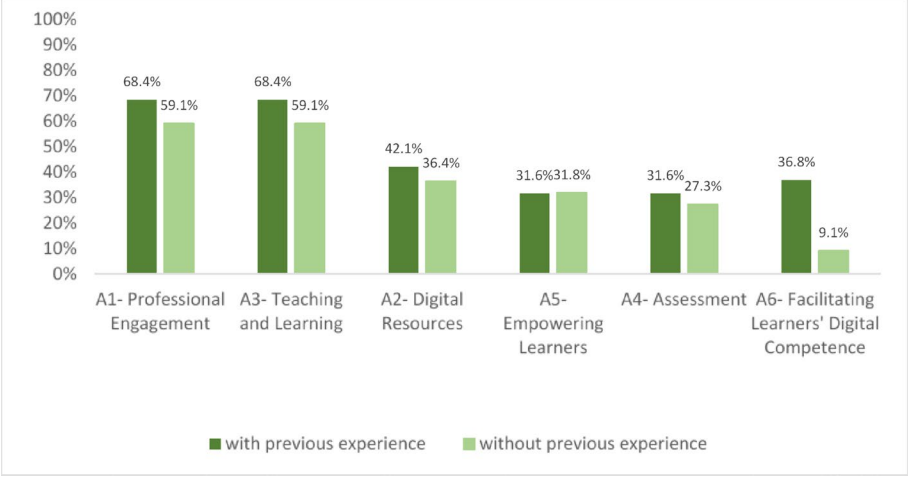


Fig. 3 Percentage of participants mentioning each area by previous experience

[5] We present these findings as percentages because the sample sizes of the two groups are different, and percentages allow for a meaningful comparison of the relative importance or prevalence of each area irrespective of their absolute numbers.

Table 2 Number of participants mentioning each specific competence (sorted by area) (It is important to note that the participants could mention multiple competences within a single area. Consequently, the total number of mentions across all areas may be larger than the figures presented in Fig. 1.)

Professional Engagement	Organisational communication	23
	Professional collaboration	9
	Reflective practice	0
	Digital CPD	10
Digital Resources	Selecting	15
	Creating and modifying	7
	Managing, protecting, sharing	0
Teaching & Learning	Teaching	24
	Guidance	2
	Collaborative learning	4
	Self-regulated learning	2
Assessment	Assessment strategies	12
	Analysisng evidence	1
	Feedback & planning	2
Empowering Learners	Differentiation & personalisation	1
	Accessibility & inclusion	7
	Actively engaging learners	6
Facilitating Learners' Digital Competence	Information & media literacy	2
	Communication	3
	Content creation	4
	Responsible use	4
	Problem solving	0

Professional engagement

The area of "Professional Engagement" of DigCompEdu was the most relevant to the participants of the study. In this scope, the "Organizational Communication" competence emerged as the most highlighted (n = 23; 56.1% of the total), underlining the importance that the participants granted to digital technologies to facilitate communication among teachers, students and families. Particularly, the participants considered that improving communication with families is a fundamental pillar in the use of ICT. This perspective

towards the use of ICT shows the importance of collaboration and the creation of relationships between the families and the school, since they can result in greater student engagement. In their responses, the participants identified different digital tools that facilitate communication with the families, such as online meetings, blogs, social media, tutorial management systems, and platforms to gather inquiries from the families or to report non-attendance.

The "Professional Engagement" area of DigCompEdu was the most relevant to participants. "Organisational Communication" stood out (n = 23; 56.1%), emphasising the role of digital technologies in facilitating communication among teachers, students, and families. Participants viewed improving family communication as a key ICT function, fostering collaboration and stronger school-family relationships, which can enhance student engagement. They identified various digital tools for this purpose, including online meetings, blogs, social media, tutorial management systems, and platforms for enquiries and attendance reporting.

Another two competences that were mentioned by the participants in this area were "Digital CPD" (n = 10; 24.4%) and "Professional Collaboration" (n = 9; 22%). In regard with the former, the participants valued the possibility of accessing online courses, as well as conferences and videos for both "using digital resources and training in other aspects" (Document 8; woman, 22 years old), underlining the flexibility offered by this modality. In relation to "Professional Collaboration", the participants also regarded this competence as a source of professional development, and they highlighted that it is especially strengthened by social networks, enabling the exchange of good practices, "experiences, and training resources among educators" (Document 7; woman, 27 years old). The participants recognized the value of technologies to establish means of communication, cooperation and collaboration (Document 5; woman, 43 years old), beyond the school itself, thus enriching the teaching practice and promoting educational innovation.

Participants also mentioned "Digital CPD" (n = 10; 24.4%) and "Professional Collaboration" (n = 9; 22%). Regarding the former, they valued access to online courses, conferences, and videos for both "using digital resources and training in other aspects" (Document 8; woman, 22), highlighting the flexibility of this format. As for "Professional Collaboration", they saw it as a means of professional development, particularly enhanced by social networks, which facilitate sharing good practices, "experiences, and training resources among educators" (Document 7; woman, 27). Participants recognised the role of technology in fostering communication, cooperation, and collaboration (Document 5; woman, 43) beyond the school itself, enriching teaching practice and promoting educational innovation.

Lastly, it is important to highlight that "Reflective Practice" was not pointed out by any of the participants in the area of "Professional Engagement". This competence refers to the capacity of educators to critically evaluate the use of digital technologies in their professional practice, both their own and that of their colleagues.

Teaching and learning

"Teaching and Learning" was the second most mentioned area, largely due to the prominence of the "Teaching" competence (n = 24; 58.5%). Participants highlighted the effectiveness of teaching methodologies, the ability to convey knowledge clearly and engagingly, and, above all, the importance of connecting with and motivating students.

An important aspect of 'Teaching' was the emphasis on gamification as a strategy to enhance student motivation: "Teaching through games in the classroom making use of technologies is also a good resource to help students familiarize with ICT; they learn things related to the subject matter and they constitute a much more attractive method for the students (Document 41; woman, 24). Participants valued digital technologies for creating engaging learning experiences through educational games and reward systems.

On the other hand, the rest of the specific competences of this area received poor attention in the responses of the participants. The "Collaborative Learning" competence was only mentioned by four participants (9.8%), from an approach focused on the use of digital tools to share ideas, without delving into the development of specific didactic methodologies that promote collaboration, for example: "Using tools such as Padlet to promote the use of digital applications that allow sharing the ideas of the students" (Document 9; woman, 23 years old).

The other specific competences in this area received little attention. "Collaborative Learning" was mentioned by only four participants (9.8%), with a focus on using digital tools to share ideas, rather than developing specific teaching methodologies for collaboration. For example: "Using tools such as Padlet to promote the use of digital applications that allow sharing the ideas of the students" (Document 9; woman, 23)."

Lastly, the possibilities offered by digital technologies to provide new ways of guiding and supporting learning (Guidance) were rarely mentioned (n = 2; 4.9% of the total). However, the qualitative data show some examples that highlight, especially, uses of ICT that allow offering a better scaffolding for learning, such as "videotutorials and tools to perform schedules" or "already solved exercises and videos that show the processes" for the solution as a guide.

Digital resources

"Digital Resources" was the third most mentioned area, showing a moderate importance. Within this area, the specific competence of "Selection of digital resources that already exist in the network" stood out (n = 15; 36.6%). The mentions to the selection of digital resources were not limited to the search of static content for use in the classroom. The participants also valued the creation of multimodal presentations and the inclusion of more updated content that may support methodological changes. An example of this is the selection of applications or games "to learn and have a more significant learning" (Document 6; man, 33).

The competence of "Creating and Modifying Digital Resources" was mentioned by fewer participants (n = 7; 17.1%) than mere selection. The responses in this area highlight the value of digital technologies in the creation of ad-hoc materials to "work on aspects that they consider of great use for the class" (Document 17; man, 25), adapting them to the specific needs of the students and their context.

Curiously, no participant mentioned the specific competence of "Managing, Protecting and Sharing Digital Resources", which involves technical, legal, ethical, and security aspects in managing digital resources. This absence is particularly significant among future school principals.

Empowering learners

"Empowering Learners" was not frequently mentioned by the participants as an area of teacher digital competence. However, the specific competence of "Accessibility and Inclusion" (n = 7, 17.1%) stood out. These participants agreed that teachers must master digital technologies to support educational inclusion, addressing student diversity. This implies considering "Special Educational Needs" (Document 10; woman, 31) and using specific tools for challenges. For example, participants mentioned "translators or screen readers" (Document 38; woman, 26) to facilitate access to information for migrant students or students with visual impairment; reading-support tools, such as tools for the "identification of syllables and sounds" (Reflection Exercise 3; woman, 22) for students with dyslexia or reading problems; and online education for students unable to attend in person.

In this study, the "Actively Engaging Learners" competence was only mentioned by 6 participants (n = 6), who underlined the potential of digital technologies to stimulate creativity and research. They valued methodologies that encourage students to create online content, research real-world problems using the Internet, and experiment with "virtual reality and simulations".

Lastly, the "Differentiation and Personalization" competence was mentioned once (n = 1, 2.4%), highlighting ICT's capacity to adapt learning to individual needs, enabling students to take quick, customised tests and "personalise learning, with the aim of adapting the content and learning pace to the individual needs of the students" (Document 32; woman, 22).

Assessment

Assessment was another area with few mentions, primarily focused on the specific competence of "Assessment Strategies" (n = 12; 29.3%).

Regarding the "Assessment Strategies", participants highlighted the value of digital technologies in diversifying assessment methodologies. One participant commented: "all this scaffolding we mention can be assessed. Therefore, there are different technological tools that evaluate the entire academic development of the student and allows oneself to be original in his/her own evaluation" (Document 10; woman, 22). Similarly, the participants underlined the capacity of technologies to conduct a more individualized assessment: "since there are tools that facilitate and promote this task" (Document 2; woman, 27).

However, mentions related to improving evaluation processes through "Analysing Evidence" (n = 1; 2.4%) and better "Feedback and Planning" (n = 2; 4.9%) were rare.

The only mention of evidence analysis focused on the ability of digital technologies to gather real-time information on students' academic performance. It was stated that "digital technologies can gather information about the students at any time of the day" (Document 16; woman, 22), either through automatically generated data or teacher observations. Similarly, mentions about improving feedback centred on technologies that provide useful data "to evaluate the students, observe their progress and offer them adequate feedback" (Document 32; woman, 22).

Facilitating learners' digital competence

The study reveals that "Facilitating Learners' Digital Competence" was the least mentioned area. Despite general recognition of its importance, reflected in comments such as "future students will work with technologies" (Document 9; woman, 23), the mention of specific competence in this area was very low. Moreover, it was predominantly mentioned by participants with prior professional experience, rather than being equally considered across all groups.

The few mentions to the "Content Creation" competence (n = 4, 9.8%) focused on the development of contents using ICT, such as creating classroom blogs, for example: "The creation of a classroom blog managed by the students, where we can upload everything we do in class" (Document 4; woman, 23).

Mentions of the responsible use of technology were also scarce (n = 4, 9.8%), focusing on the importance of promoting a responsible use from the schools, with impact beyond academics. A participant commented: "The digital learning of the students is essential nowadays, both in content and in responsible use, thus the daily use of ICT in the classroom must favor it" (Document 5; woman, 43).

There were also few mentions of the "students´ communication" (n = 3, 7.3%), with participants highlighting the potential of technologies for simultaneous and joint work on academic documents "both inside and outside of the classroom" (Document 6; man, 33), although without delving into the transfer of these skills beyond the academic context.

The last element mentioned was the development of "Information and Media Literacy" (n = 2, 4.9%), where participants focused on the autonomous search for information using various online resources, addressing advanced aspects like source verification or information authenticity (e.g., "to ensure that children can search for information on their own with the resources they have today in the Internet, such as Google Maps, Wikipedia, Translators, etc." (Document 39; woman, 24).

Finally, it is noteworthy that the "Problem Solving" competence was not mentioned. This competence includes not only technical aspects but also the ability to apply technological knowledge to creatively address new situations in everyday life.

Discussion and conclusion

The use of digital technologies to improve education is an aspect of social and educational relevance (OECD, 2015; UNESCO, 2023). Therefore, in the last years, organizations, policymakers and researchers have defined the digital competence that educators must have to improve their professional practice (European Commission, 2020; INTEF, 2017; Redecker & Punie, 2017; UNESCO, 2011).

Digital competence is not limited to teachers, as it must be an attribute of all education professionals, especially school leaders (Caena & Redecker, 2019; Pettersson, 2018). The latter play an important role in the promotion of a pedagogical and technological type of leadership, promoting the improvement of the teaching and learning processes through digital technologies (Anderson & Dexter, 2005). However, it is unclear whether teachers are receiving enough training to contemplate all the aspects of teacher digital competence. To respond to this question, a study was carried out with two cohorts (academic year 22/23 and 23/24) of students of a Spanish master's degree in school management.

In this study, the students were asked to reflect on how teachers can use technologies in their professional practice (teacher digital competence) openly and without guidance.

Their responses were compared to the 22 competences defined by the European framework of digital competences for educators (DigCompEdu). The authors assumed that the areas and competences of the framework that were least mentioned pose gaps in their view of the professional digital competence of teachers and, on the contrary, the aspects that the participants have internalized to a greater extent.

The results reveal that the participants do not have a holistic view of the digital competence that teachers must possess (Instefjord & Munthe, 2017; Pettersson, 2018; Tondeur et al., 2017). Only two areas, "Professional Engagement" and "Teaching and Learning", were mentioned by more than half of the participants. Thus, the rest of the areas appear to be less covered in the training plans, especially the development of students' digital competence, which is an outcome-oriented rather than process-oriented area.

Moreover, the analysis of specific competences showed a superficial view in many areas, demonstrating the need for further training. In "Organizational Communication", although the students recognized the improvement in communication with different educational agents, they omitted the reflective practice on the use of technologies, which is a key competence for a responsible pedagogical and digital leadership. In "Teaching and Learning", the approach was focused traditional teaching, whereas advanced aspects of the use of technologies such as collaborative learning and support to self-regulated learning were little internalized. Areas with fewer mentions, such as "Digital Resources", were predominated by basic competences (e.g., search and selection), whereas the creation, management and protection of digital resources were little considered. The same was observed in "Assessment", where advanced aspects such as digital evidence analysis and the use of technologies for relevant feedback were rarely mentioned. In "Empowering Learners", "Differentiation and Personalization", which are essential to attend to student diversity (Yasar-Akyar et al., 2022), were not mentioned. Lastly, the development of student digital competence, which is a key area integrated in the curriculum (European Commission, 2019), was not strongly associated by the participants with teacher digital competence.

Our results are a contribution to a field of research that has been poorly studied, i.e., the conception of teacher digital competence in future school leaders, and their importance lies in four key aspects. Firstly, they reveal possible causes of difficulties that principals must face to manage the digital transformation of schools, which have been identified in previous studies (Flanagan & Jacobsen, 2003; Olofsson et al., 2015). In this sense, disregarding many of the specific competences that make up teacher digital competence may hinder the guidance towards an effective use of technologies. Secondly, they describe the conception of future principals about teacher digital competence. Thirdly, they offer an early view of the situation, as they evaluate students who have not yet taken the role of school leaders but are training for it, which allows identifying the formative divides from the beginning and provides insights to coordinate and optimize initial and in-service training. Lastly, they demonstrate the usefulness of employing competence frameworks, such as DigCompEdu, in research, making the latter more reproducible and comparable (Haşlaman et al., 2024), as well as in mapping contents, identifying gaps and providing guidelines on how to update the training plans of school leaders. To sum up, the study offers an empirical foundation to discuss the training of future school leaders in digital competence.

This study provides insights into the understanding and perception of teacher digital competence among future school leaders. However, it's essential to acknowledge some limitations. Methodologically, an interpretivist approach was employed, focusing on a specific case within a particular training program. While this restricts the direct extrapolation of results to educational contexts, the methodological design allows for transferability. Indeed, multiple recent studies in diverse contexts (Berkovich & Hassan, 2023; Chen &and Kouhsari, 2025; Kafa, 2025; Lozano-Díaz et al., 2024) support the need to strengthen digital leadership training, confirming the relevance of an approach like the one used here to understand better the strengths and weakness. Regarding the generalization of the results, although we focus on a specific program, insights gained may be similar and transferable to similar populations or educational settings, always recognizing the influence of contextual factors. Furthermore, while the results are robust among students with and without professional experience as teachers**,** the participants did not play the role of school leaders when the study was conducted, thus their view on teacher digital competence could evolve with experience and in-service school leader training. In the future, it is recommended to replicate the study with active principals and compare the results to obtain a more complete perspective.

The main conclusion of our study is that, in spite of having recent training, the future educational leaders who participated in the study did not know all the dimensions of teacher digital competence, as defined by DigCompEdu. Therefore, the attainment of educational digital transformation requires a holistic training that goes beyond superficial aspects about the possibilities that digital technologies offer to (future) principals. Adopting this strategy can enhance the development of pedagogical and digital leadership in line with modern leadership theories and the societal requirements.

Since the development of students' digital competence is already integrated into the curricula of many countries (EACEA, 2023), the key challenge lies in promoting the use of technology for other, less mandatory pedagogical issues. To achieve this, the DigCompEdu framework can be explicitly adopted as a roadmap for advancing techno-pedagogical leadership and identifying training needs, not just for teachers (as done in some countries), but also for school leaders. For instance, in our case, we propose a starting point that emphasizes awareness and knowledge among future leaders on how to use digital technologies for inclusive and personalized learning (Empowering Learners) and to diversify and improve assessment practices as core dimensions. Furthermore, if digitalisation of schools is a priority, to be more comprehensive and ensure accountability and strategic focus, it could be appropriate to review the criteria for access to and permanence in leadership positions considering elements that combine digital and pedagogical leadership. This review should ensure that the Educational Inspectorate explicitly assesses the areas of the DigCompEdu framework within school leadership projects and principals' practices. Only by aligning training and policy in this way it will be possible to foster leadership that coherently integrates pedagogical and technological dimensions for the improvement of educational institutions.

ANNEX I: Book of codes based on the DigCompEdu framework

Category	Subcategory	Codes	Definition
Professional competences of educators	1. Professional Engagement	Organizational Communication	Use digital technologies to improve organizational communication with students, parents and third parties Contribute to the development and improvement of organizational communication strategies
		Professional Collaboration	Use digital technologies to collaborate with other educators, sharing knowledge and experiences; innovating pedagogical practices in a collaborative manner Use collaborative networks of professionals as a source of professional development
		Reflective Practice	Reflect, critically evaluate and actively develop their own digital pedagogical practice and that of their educational community
		Digital CPD	Use digital sources and resources for continuing professional development
Pedagogical competences of educators	2. Digital Resources	Selecting	Identify, evaluate and select digital resources for teaching and learning, understand applicable copyright and accessibility requirements Modify existing openly licensed resources and other resources where permitted
		Creating & Modifying	Create or co-create new digital educational resources Consider the specific learning objective, context, pedagogical approach and learner group when designing digital resources and planning their use
		Managing, protecting, sharing	Organize digital content and make it available to students, parents, or other educators Effectively protect sensitive digital content. Respect privacy and copyright regulations Understand the use and creation of open licenses and open educational resources, including proper attribution
	3. Teaching and Learning	Teaching	Integrate digital devices and resources into the teaching process in order to improve the effectiveness of teaching practices Adequately adapt the bases, manage and orchestrate digital teaching interventions Experiment and develop new formats and pedagogical methods of instruction
		Guidance	Use digital tools and services to enhance interaction with students, individually and collectively, inside and outside the learning session Use digital technologies to provide timely and targeted guidance and assistance Experiment with and develop new ways and formats to offer guidance and support
		Collaborative Learning	Use digital technologies to foster and enhance collaborative learning strategies, e.g., as a basis for collaborative group sharing, as a tool for conducting a collaborative assignment, or as a means of presenting results
		Self-regulated Learning	Use digital technologies to support self-directed learning processes, i.e., to enable students to plan, monitor and reflect on their own learning, evidence progress, share knowledge and present creative solutions
	4. Assessment	Assessment Strategies	Use digital tools for formative and summative assessment Improve the diversity and appropriateness of assessment formats and approaches

Category	Subcategory	Codes	Definition
		Analysing Evidence	Generate, select, critically analyze, and interpret digital evidence of student digital activity, performance, and progress in order to inform teaching and learning
		Feedback & Planning	Use digital tools to provide timely feedback to students Appropriately adapt teaching strategies and provide targeted support based on evidence generated by the tools used Help students and parents understand the evidence provided by digital tools and use them for decision making
	5. Empowering Learners	Accesibility & Inclusion	Ensure accessibility to learning resources and activities for all learners, including those with special needs Consider and respond to learners' digital expectations, skills, digital uses and misconceptions, as well as contextual, physical or cognitive constraints to the use of digital tools
		Differentiation & Personalization	Use digital tools to address the diverse learning needs of students, e.g., allowing them to pursue different learning paths and goals, offering alternative approaches and tools, and enabling students to move at different speeds toward individual learning goals
		Actively Engaging Learners	Use digital technologies to foster students' cross-cutting skills and creative expression Open learning to real-world contexts, engage students in hands-on activities, scientific inquiry, complex problem solving, creative expression
Student competences	6. Development of Student Digital Competence	Information & Media Literacy	Incorporate learning activities, tasks/assignments, and assessments that require students to articulate information needs; find information and resources in digital environments; organize, process, analyze, and interpret information; and compare and critically evaluate the credibility and reliability of information and its sources
		Communication	Incorporate learning activities, assignments, and assessments that require students to effectively and responsibly use digital tools for communication, collaboration, and civic engagement Incorporate learning activities and assignments that require students to express themselves through digital media and to modify and create digital content in different formats
		Content Creation	Teach students how copyright and licensing apply to digital content, how to reference sources and apply licenses
		Responsible Use	Take measures to ensure the physical, psychological and social well-being of students when using digital technologies Empower students to manage risks and use digital technologies to support their own social, psychological and physical well-being
		Problem Solving	Incorporate learning and assessment activities that require students to identify and solve technical problems or creatively transfer their technological knowledge to new situations

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Data availability
An anonymized version of the data used and analysed during the current study is available from the corresponding author on reasonable request.

Declarations

Competing interests
None.

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References

Acton, K. S. (2021). School leaders as change agents: Do principals have the tools they need? *Management in Education, 35*(1), 43–51. https://doi.org/10.1177/0892020620927415
Afshari, M., Bakar, K. A., Luan, W. S., & Siraj, S. (2012). Factors affecting the transformational leadership role of principals in implementing ICT in schools. *Turkish Online Journal of Educational Technology - TOJET, 11*(4), 164–176.
Agyei, D. D., & Voogt, J. (2014). Examining factors affecting beginning teachers' transfer of learning of ICT-enhanced learning activities in their teaching practice. *Australasian Journal of Educational Technology*. https://doi.org/10.14742/ajet.499
Ahmad, F., Jhajj, A. K., Stewart, D. E., Burghardt, M., & Bierman, A. S. (2014). Single item measures of self-rated mental health: A scoping review. *BMC Health Services Research, 14*(1), Article 398. https://doi.org/10.1186/1472-6963-14-398
Allen, M. S., Iliescu, D., & Greiff, S. (2022). Single item measures in psychological science: A call to action. *European Journal of Psychological Assessment, 38*(1), 1–5. https://doi.org/10.1027/1015-5759/a000699
Anderson, R. E., & Dexter, S. (2005). School Technology Leadership: An empirical investigation of prevalence and effect. *Educational Administration Quarterly, 41*(1), 49–82. https://doi.org/10.1177/0013161X04269517
Ang, L., & Eisend, M. (2018). Single versus multiple measurement of attitudes: A meta-analysis of advertising studies validates the single-item measure approach. *Journal of Advertising Research, 58*(2), 218–227. https://doi.org/10.2501/JAR-2017-001
Avidov-Ungar, O., Shamir-Inbal, T., & Blau, I. (2022). Typology of digital leadership roles tasked with integrating new technologies into teaching: Insights from metaphor analysis. *Journal of Research on Technology in Education, 54*(1), 92–107. https://doi.org/10.1080/15391523.2020.1809035
Bastos, G., & Oliveira, I. (2015). Digital Literacy of School Leaders: what impacts in schools? Results of two studies from Portugal. In *Information Literacy: Moving Toward Sustainability: Third European Conference, ECIL 2015, Tallinn, Estonia, October 19-22, 2015, Revised Selected Papers 3* (pp. 132–142). Springer International Publishing.
Berkovich, I., & Hassan, T. (2023). Principals' digital transformational leadership, teachers' commitment, and school effectiveness. *Education Inquiry*. https://doi.org/10.1080/20004508.2023.2173705
Bernard, R. M., Abrami, P. C., Borokhovski, E., Wade, C. A., Tamim, R. M., Surkes, M. A., & Bethel, E. C. (2009). A meta-analysis of three types of interaction treatments in distance education. *Review of Educational Research, 79*(3), 1243–1289. https://doi.org/10.3102/0034654309333844
Blau, I., & Shamir-Inbal, T. (2017). Digital competences and long-term ICT integration in school culture: The perspective of elementary school leaders. *Education and Information Technologies, 22*, 769–787. https://doi.org/10.1007/S10639-015-9456-7
Caena, F., & Redecker, C. (2019). Aligning teacher competence frameworks to 21st century challenges: The case for the European Digital Competence Framework for Educators (*digcompedu). European Journal of Education, 54*(3), 356–369. https://doi.org/10.1111/ejed.12345
Castaño Muñoz, J., Vuorikari, R., Costa, P., Hippe, R., & Kampylis, P. (2021). Teacher collaboration and students' digital competence - Evidence from the SELFIE tool. *European Journal of Teacher Education*. https://doi.org/10.1080/02619768.2021.1938535
Castaño-Muñoz, J., Duart, J. M., & Sancho-Vinuesa, T. (2014). The internet in face-to-face higher education: Can interactive learning improve academic achievement? *British Journal of Educational Technology*. https://doi.org/10.1111/bjet.12007
Chang, I.-H. (2012). The effect of principals' technological leadership on teachers' technological literacy and teaching effectiveness in Taiwanese elementary schools. *Journal of Educational Technology & Society, 15*(2), 328–340.
Chen, J., & Kouhsari, M. (2025). Demystifying the personal and social forces behind school digital transformation: An analysis of the job demands and resources theory through school leaders. *Computers and Education*. https://doi.org/10.1016/j.compedu.2024.105232
Cheng, L., Ritzhaupt, A. D., & Antonenko, P. (2019). Effects of the flipped classroom instructional strategy on students' learning outcomes: A meta-analysis. *Educational Technology Research and Development, 67*(4), 793–824. https://doi.org/10.1007/S11423-018-9633-7
Costa, P., Castaño-Muñoz, J., & Kampylis, P. (2021). Capturing schools' digital capacity: Psychometric analyses of the SELFIE self-reflection tool. *Computers and Education*. https://doi.org/10.1016/j.compedu.2020.104080
Day, C., Sammons, P., & Gorgen, K. (2020). Successful School Leadership. *Education development trust*.
Drisko, J. W., & Maschi, T. (2016). *Content analysis* (p. 208). Oxford University.
EACEA (2023). *Structural indicators for monitoring education and training systems in Europe 2023: Digital competence at school*. Publications Office of the European Union. https://doi.org/10.2797/886074.
Erikson, M., Erikson, M. G., & Punzi, E. (2018). A single-question qualitative bachelor's programme evaluation. *Assessment & Evaluation in Higher Education, 43*(6), 969–978. https://doi.org/10.1080/02602938.2018.1424319
Escueta, M., Nickow, A. J., Oreopoulos, P., & Quan, V. (2020). Upgrading education with technology: Insights from experimental research. *Journal of Economic Literature, 58*(4), 897–996. https://doi.org/10.1257/jel.20191507

European Commission (2019). Eurydice 2019: Digital education at school in Europe. *Education, Audiovisual and Culture Executive Agency, 2019*, 1–147. https://doi.org/10.2797/763
European Commission (2020). *Digital Education Action Plan 2021–2027: Resetting Education and Training for the Digital Age.* Retrieved from: https://ec.europa.eu/education/education-in-the-eu/digital-education-action-plan_en
Flanagan, L., & Jacobsen, M. (2003). Technology leadership for the twenty-first century principal. *Journal of Educational Administration, 41*(2), 124–142. https://doi.org/10.1108/09578230310464648
Fuchs, C., & Diamantopoulos, A. (2009). Using single-item measures for construct measurement in management research: Conceptual issues and application guidelines. *Die Betriebswirtschaft, 69*(2), 195–210.
Fyfe, E. R. (2016). Providing feedback on computer-based algebra homework in middle-school classrooms. *Computers in Human Behavior, 63*, 568–574. https://doi.org/10.1016/J.CHB.2016.05.082
Glatthorn, A. A., & Jailall, J. M. (2009). *The principal as curriculum leader: Shaping what is taught and tested*. Corwin Press.
Håkansson Lindqvist, M. (2019). School leaders' practices for innovative use of digital technologies in schools. *British Journal of Educational Technology, 50*(3), 1226–1240. https://doi.org/10.1111/BJET.12782
Haşlaman, T., Atman Uslu, N., & Mumcu, F. (2024). Development and in-depth investigation of pre-service teachers' digital competencies based on DigCompEdu: A case study. *Quality & Quantity, 58*(1), 961–986. https://doi.org/10.1007/s11135-023-01674-z
Hattie, J., & Timperley, H. (2007). The Power of Feedback. *Review of Educational Research, 77*(1), 81–112. https://doi.org/10.3102/003465430298487
Holloway, W., Jefferson, T., & Hollway, W. (2008). The free association narrative interview method. In L. Given (Ed.), *The SAGE Encyclopedia of Qualitative Research Methods.* (pp. 296–315).
Hrustek, N. Ž., Mekovec, R., & Kirinić, V. (2023). *E-Schools project-framework for digital competencies of school principals: Competencies in the area digital infrastructure.*
Instefjord, E. J., & Munthe, E. (2017). Educating digitally competent teachers: A study of integration of professional digital competence in teacher education. *Teaching and Teacher Education, 67*, 37–45. https://doi.org/10.1016/j.tate.2017.05.016
INTEF (2017). Marco Común de Competencia Digital Docente. Instituto Nacional de Tecnologías Educativas y de Formación del Profesorado. Retrieved from: https://intef.es/Noticias/el-marco-comun-de-competencia-digital-docente
ISTE [International Society for Technology in Education]. (2000). *ISTE national educational technology standards (NETS) and performance indicators for teachers*. Retrieved from: https://www.hbgdiocese.org/wp-content/uploads/2012/04/NETS-for-Teachers.pdf
Johnson, D., & Mielke, N. (2013). *Rubric for effective teacher technology use (organized by the four domains of Danielson's framework for teaching 1)*.
Kafa, A. (2025). Exploring integration aspects of school leadership in the context of digitalization and artificial intelligence. *International Journal of Educational Management, 39*(8), 98–115. https://doi.org/10.1108/IJEM-11-2024-0703
Kampylis, P., Punie, Y., & Devine, J. (2015). *Promoting effective digital-age learning: A European framework for digitally competent educational organisations*. Publications Office of the European Union. https://doi.org/10.2791/54070.
Karakose, T., Polat, H., Tülübaş, T., & Demirkol, M. (2024). A review of the conceptual structure and evolution of digital leadership research in education. *Education Sciences, 14*(11), Article 1166. https://doi.org/10.3390/educsci14111166
Kelentrić, M., Helland, K., & Arstorp, A. T. (2017). Professional digital competence framework for teachers. *The Norwegian Centre for ICT in Education, 134*(1), 1–74.
Krein, U. (2025). Exploring the transformative impact of technology on school leadership. Affordances, practices, implications. *Leadership and Policy in Schools, 24*(1), 90–102. https://doi.org/10.1080/15700763.2024.2440902
Lozano-Díaz, A., Fernández-Prados, J. S., & González-Martín, B. (2024). From crisis leadership to digital and inclusive leadership in the aftermath of the pandemic. *School Leadership & Management, 44*(4), 373–388. https://doi.org/10.1080/13632434.2024.2367985
Lucas, M., Bem-Haja, P., Siddiq, F., Moreira, A., & Redecker, C. (2021). The relation between in-service teachers' digital competence and personal and contextual factors: What matters most? *Computers & Education, 160*, Article 104052. https://doi.org/10.1016/J.COMPEDU.2020.104052
Makri, A., & Vlachopoulos, D. (2019). Professional development for school leaders: A focus on soft and digital skills. *EDULEARN19 Proceedings, 1*, 6200–6209. https://doi.org/10.21125/EDULEARN.2019.1492
Masoumi, D., & Noroozi, O. (2023). Developing early career teachers' professional digital competence: A systematic literature review. *European Journal of Teacher Education*. https://doi.org/10.1080/02619768.2023.2229006
McLaren, B. M., Van Gog, T., Ganoe, C., Karabinos, M., & Yaron, D. (2016). The efficiency of worked examples compared to erroneous examples, tutored problem solving, and problem solving in computer-based learning environments. *Computers in Human Behavior, 55*, 87–99. https://doi.org/10.1016/J.CHB.2015.08.038
MENTEP (2018). TET-SAT. Practical Guidelines for Teachers. Retrieved from: http://mentep.eun.org/documents/2390578/2487445/MENTEP_TETSAT-Guide-teachers.pdf/3c32a8a5-79cd-49b2-8ecf-420d2636228c
Mishra, P., & Koehler, M. J. (2006). Technological pedagogical content knowledge: A framework for teacher knowledge. *Teachers College Record, 108*(6), 1017–1054.
Moreno, R., & Mayer, R. (2007). Interactive multimodal learning environments: Special issue on interactive learning environments: Contemporary issues and trends. *Educational Psychology Review, 19*(3), 309–326. https://doi.org/10.1007/S10648-007-9047-2
OECD. (2015). Students, computers and learning: Making the connection. *OECD Publishing*. https://doi.org/10.1787/9789264239555-en
Olofsson, A. D., Ola Lindberg, J., Fransson, G., & Hauge, T. E. (2015). Uptake and use of digital technologies in primary and secondary schools - A thematic review of research. *Nordic Journal of Digital Literacy, 2015*(4), 103–121. https://doi.org/10.18261/ISSN1891-943X-2015-JUBILEUMSNUMMER-08
Perrotta, C. (2013). Do school-level factors influence the educational benefits of digital technology? A critical analysis of teachers' perceptions. *British Journal of Educational Technology, 44*(2), 314–327. https://doi.org/10.1111/J.1467-8535.2012.01304.X
Pettersson, F. (2018). On the issues of digital competence in educational contexts – A review of literature. *Education and Information Technologies, 23*(3), 1005–1021. https://doi.org/10.1007/s10639-017-9649-3
Pont, B., Nusche, D., & Moorman, H. (2008). *Improving School Leadership VOLUME 1: POLICY AND PRACTICE*. Retrieved from: https://citeseerx.ist.psu.edu/document?repid=rep1%26type=pdf%26doi=7ff08935b58a0814dd0c99fcb9103e8b294158fd

Prince, M. (2004). Does Active Learning Work? A review of the research. *Journal of Engineering Education, 93*(3), 223–231. https://doi.org/10.1002/J.2168-9830.2004.TB00809.X

Radinger, T. (2014). School leader appraisal — A tool to strengthen school leaders' pedagogical leadership and skills for teacher management? *European Journal of Education, 49*(3), 378–394. https://doi.org/10.1111/EJED.12085

Redecker, C., & Punie, Y. (Eds.). (2017). *European framework for the digital competence of educators: DigCompEdu*. Publications Office of the European Union. https://doi.org/10.2760/178382

Reis-Andersson, J. (2024). Leaders' perceptions of digitalisation in K–12 education: Influencing arrangements for leading the expansion of digital technologies. *Discover Education*. https://doi.org/10.1007/s44217-024-00247-y

Selwyn, N., Hillman, T., Eynon, R., Ferreira, G., Knox, J., Macgilchrist, F., & Sancho-Gil, J. M. (2020). What's next for Ed-Tech? Critical hopes and concerns for the 2020s. *Learning, Media and Technology, 45*(1), 1–6. https://doi.org/10.1080/17439884.2020.1694945

Surma, T., & Kirschner, P. A. (2020). Technology enhanced distance learning should not forget how learning happens. *Computers in Human Behavior, 110*, Article 106390. https://doi.org/10.1016/J.CHB.2020.106390

Timotheou, S., Miliou, O., Dimitriadis, Y., Vilagrá Sobrino, S., Giannoutsou, N., Cachia, R., Martínez Monés, A., & Ioannou, A. (2023). Impacts of digital technologies on education and factors influencing schools' digital capacity and transformation: A literature review. *Education and Information Technologies, 28*(6), 6695–6726.

Tondeur, J., Aesaert, K., Pynoo, B., van Braak, J., Fraeyman, N., & Erstad, O. (2017). Developing a validated instrument to measure preservice teachers' ICT competencies: Meeting the demands of the 21st century. *British Journal of Educational Technology, 48*(2), 462–472. https://doi.org/10.1111/bjet.12380

Tondeur, J., Cooper, M., & Newhouse, C. P. (2010). From ICT coordination to ICT integration: A longitudinal case study. *Journal of Computer Assisted Learning, 26*(4), 296–306. https://doi.org/10.1111/J.1365-2729.2010.00351.X

Tondeur, J., Forkosh-Baruch, A., Prestridge, S., Albion, P., & Edirisinghe, S. (2016). Responding to challenges in teacher professional development for ICT integration in education. *Educational Technology & Society, 19*(3), 110–120.

UNESCO (2011). UNESCO ICT competency framework for teachers. Retrieved from: https://iite.unesco.org/pics/publications/en/files/3214694.pdf

UNESCO (2023). Technology in Education: A Tool on the Rise for Sustainable Development. UNESCO Publishing. Retrieved from: https://unesdoc.unesco.org/ark:/48223/pf0000380395

Valli, L., Stefanski, A., & Jacobson, R. (2018). School-community partnership models: Implications for leadership. *International Journal of Leadership in Education, 21*(1), 31–49. https://doi.org/10.1080/13603124.2015.1124925

Vanderlinde, R., & van Braak, J. (2010). The gap between educational research and practice: Views of teachers, school leaders, intermediaries and researchers. *British Educational Research Journal, 36*, 299–316. https://doi.org/10.1080/01411920902919257

Velarde, J. M., Ghani, M. F., Adams, D., & Cheah, J. H. (2022). Towards a healthy school climate: The mediating effect of transformational leadership on cultural intelligence and organisational health. *Educational Management Administration & Leadership, 50*(1), 163–184.

Voogt, J., Fisser, P., Pareja Roblin, N., Tondeur, J., & van Braak, J. (2013). Technological pedagogical content knowledge – a review of the literature. *Journal of Computer Assisted Learning, 29*(2), 109–121. https://doi.org/10.1111/J.1365-2729.2012.00487.X

Williamson, B., Potter, J., & Eynon, R. (2019). New research problems and agendas in learning, media and technology: The editors' wishlist. *Learning, Media and Technology, 44*(2), 87–91. https://doi.org/10.1080/17439884.2019.1614953

Yasar-Akyar, O., Rosa-Feliz, C., Sunday-Oyelere, S., Muñoz, D., & Demirhan, G. (2022). Special education teacher's professional development through digital storytelling. *Comunicar, 30*(71), 1–12. https://doi.org/10.3916/C71-2022-07

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